Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z2,Z5

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Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple Z2 Z5: $\frac{Z_2 Z_5 g_m - Z_2 + Z_5}{2 Z_2 g_m + 4}$

$$H(z) = \frac{Z_2 Z_5 g_m - Z_2 + Z_5}{2 Z_2 g_m + 4}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{R_5}{C_5 R_5 s + 1},\ \infty\right)$

$$H(s) = \frac{R_5 g_m + s\left(C_2 R_5 - C_5 R_5\right) - 1}{4 C_2 C_5 R_5 s^2 + 2 g_m + s\left(4 C_2 + 2 C_5 R_5 g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_2}\sqrt{C_5}\sqrt{R_5}\sqrt{g_m}}{2C_2 + C_5 R_5 g_m}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{2\sqrt{C_2}\sqrt{C_5}\sqrt{R_5}}$
bandwidth: $\frac{2C_2 + C_5 R_5 g_m}{2 C_2 C_5 R_5}$
K-LP: $\frac{R_5 g_m - 1}{2 g_m}$
K-HP: 0
K-BP: $\frac{C_2 R_5 - C_5 R_5}{4C_2 + 2C_5 R_5 g_m}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(\infty,\ \frac{R_2}{C_2 R_2 s + 1},\ \infty,\ \infty,\ \frac{R_5}{C_5 R_5 s + 1},\ \infty\right)$

$$H(s) = \frac{R_2 R_5 g_m - R_2 + R_5 + s\left(C_2 R_2 R_5 - C_5 R_2 R_5\right)}{4 C_2 C_5 R_2 R_5 s^2 + 2 R_2 g_m + s\left(4 C_2 R_2 + 2 C_5 R_2 R_5 g_m + 4 C_5 R_5\right) + 4}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_2}\sqrt{C_5}\sqrt{R_2}\sqrt{R_5}\sqrt{R_2 g_m + 2}}{2C_2 R_2 + C_5 R_2 R_5 g_m + 2 C_5 R_5}$
wo: $\frac{\sqrt{2}\sqrt{R_2 g_m + 2}}{2\sqrt{C_2}\sqrt{C_5}\sqrt{R_2}\sqrt{R_5}}$
bandwidth: $\frac{2C_2 R_2 + C_5 R_2 R_5 g_m + 2C_5 R_5}{2 C_2 C_5 R_2 R_5}$
K-LP: $\frac{R_2 R_5 g_m - R_2 + R_5}{2 R_2 g_m + 4}$
K-HP: 0
K-BP: $\frac{C_2 R_2 R_5 - C_5 R_2 R_5}{4 C_2 R_2 + 2 C_5 R_2 R_5 g_m + 4 C_5 R_5}$
Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (\infty,\ R_2,\ \infty,\ \infty,\ R_5,\ \infty)$

$$H(s) = \frac{R_2 R_5 g_m - R_2 + R_5}{2 R_2 g_m + 4}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{-C_5 R_2 s + R_2 g_m + 1}{s\left(2 C_5 R_2 g_m + 4 C_5\right)}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ \frac{R_5}{C_5 R_5 s + 1},\ \infty\right)$

$$H(s) = \frac{-C_5 R_2 R_5 s + R_2 R_5 g_m - R_2 + R_5}{2 R_2 g_m + s\left(2 C_5 R_2 R_5 g_m + 4 C_5 R_5\right) + 4}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{R_2 g_m + s\left(C_5 R_2 R_5 g_m - C_5 R_2 + C_5 R_5\right) + 1}{s\left(2 C_5 R_2 g_m + 4 C_5\right)}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ R_5,\ \infty\right)$

$$H(s) = \frac{C_2 R_5 s + R_5 g_m - 1}{4 C_2 s + 2 g_m}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{g_m + s\left(C_2 - C_5\right)}{4 C_2 C_5 s^2 + 2 C_5 g_m s}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{C_2 C_5 R_5 s^2 + g_m + s\left(C_2 + C_5 R_5 g_m - C_5\right)}{4 C_2 C_5 s^2 + 2 C_5 g_m s}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(\infty,\ \frac{R_2}{C_2 R_2 s + 1},\ \infty,\ \infty,\ R_5,\ \infty\right)$

$$H(s) = \frac{C_2 R_2 R_5 s + R_2 R_5 g_m - R_2 + R_5}{4 C_2 R_2 s + 2 R_2 g_m + 4}$$

9.9 X-INVALID-ORDER-9 $Z(s) = \left(\infty,\ \frac{R_2}{C_2 R_2 s + 1},\ \infty,\ \infty,\ \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{R_2 g_m + s\left(C_2 R_2 - C_5 R_2\right) + 1}{4 C_2 C_5 R_2 s^2 + s\left(2 C_5 R_2 g_m + 4 C_5\right)}$$

9.10 X-INVALID-ORDER-10 $Z(s) = \left(\infty,\ \frac{R_2}{C_2 R_2 s+1},\ \infty,\ \infty,\ R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{C_2 C_5 R_2 R_5 s^2 + R_2 g_m + s\left(C_2 R_2 + C_5 R_2 R_5 g_m - C_5 R_2 + C_5 R_5\right) + 1}{4 C_2 C_5 R_2 s^2 + s\left(2 C_5 R_2 g_m + 4 C_5\right)}$$

9.11 X-INVALID-ORDER-11 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ R_5,\ \infty\right)$

$$H(s) = \frac{R_5 g_m + s\left(C_2 R_2 R_5 g_m - C_2 R_2 + C_2 R_5\right) - 1}{2 g_m + s\left(2 C_2 R_2 g_m + 4 C_2\right)}$$

9.12 X-INVALID-ORDER-12 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{-C_2 C_5 R_2 s^2 + g_m + s\left(C_2 R_2 g_m + C_2 - C_5\right)}{2 C_5 g_m s + s^2\left(2 C_2 C_5 R_2 g_m + 4 C_2 C_5\right)}$$

9.13 X-INVALID-ORDER-13 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{g_m + s^2\left(C_2 C_5 R_2 R_5 g_m - C_2 C_5 R_2 + C_2 C_5 R_5\right) + s\left(C_2 R_2 g_m + C_2 + C_5 R_5 g_m - C_5\right)}{2 C_5 g_m s + s^2\left(2 C_2 C_5 R_2 g_m + 4 C_2 C_5\right)}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \frac{R_5}{C_5 R_5 s+1},\ \infty\right)$

$$H(s) = \frac{-C_2 C_5 R_2 R_5 s^2 + R_5 g_m + s\left(C_2 R_2 R_5 g_m - C_2 R_2 + C_2 R_5 - C_5 R_5\right) - 1}{2 g_m + s^2\left(2 C_2 C_5 R_2 R_5 g_m + 4 C_2 C_5 R_5\right) + s\left(2 C_2 R_2 g_m + 4 C_2 + 2 C_5 R_5 g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_2}\sqrt{C_5} R_2 \sqrt{R_5} g_m \sqrt{\frac{g_m}{R_2 g_m+2}} + 2\sqrt{C_2}\sqrt{C_5}\sqrt{R_5}\sqrt{\frac{g_m}{R_2 g_m+2}}}{C_2 R_2 g_m + 2 C_2 + C_5 R_5 g_m}$

wo: $\sqrt{\frac{g_m}{C_2 C_5 R_2 R_5 g_m + 2 C_2 C_5 R_5}}$

bandwidth: $\frac{\sqrt{\frac{g_m}{C_2 C_5 R_2 R_5 g_m + 2 C_2 C_5 R_5}}\left(C_2 R_2 g_m + 2 C_2 + C_5 R_5 g_m\right)}{\sqrt{C_2}\sqrt{C_5} R_2 \sqrt{R_5} g_m \sqrt{\frac{g_m}{R_2 g_m+2}} + 2\sqrt{C_2}\sqrt{C_5}\sqrt{R_5}\sqrt{\frac{g_m}{R_2 g_m+2}}}$

K-LP: $\frac{R_5 g_m - 1}{2 g_m}$

K-HP: $-\frac{R_2}{2 R_2 g_m + 4}$

K-BP: $\frac{C_2 R_2 R_5 g_m - C_2 R_2 + C_2 R_5 - C_5 R_5}{2 C_2 R_2 g_m + 4 C_2 + 2 C_5 R_5 g_m}$

Qz: None

Wz: $\frac{\sqrt{-R_5 g_m + 1}}{\sqrt{C_2}\sqrt{C_5}\sqrt{R_2}\sqrt{R_5}}$

11 X-PolynomialError